

Cellular Automata as Non-Associative Algebraic Fields:

The Sedenion Origin of Rule 30 Chaos and the Emergence of Geometric Stiffness

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Abstract

We propose a fundamental isomorphism between the Class 3 chaotic behavior of Elementary Cellular Automata (specifically Rule 30) and the proliferation of zero divisors in non-associative Cayley-Dickson algebras. Within the Axiomatic Physical Homeostasis (APH) framework, we re-interpret the cellular lattice not as a computational grid, but as a discrete algebraic field. We demonstrate that the pseudorandom central column of Rule 30 corresponds to the *Associator Anomaly* of the Sedenion algebra ($\mathbb{S}$). Furthermore, we introduce the **Geometric Stiffness Parameter** (β) into the update rule. We show that Rule 30 represents the limit $\beta \rightarrow 0$ (Zero Viscosity/Sedenion Phase), while the Standard Model of particle physics emerges in the stiff limit $\beta \rightarrow 1.91$ (Octonion Phase), where algebraic error-correction suppresses the chaotic propagation.

1 Introduction: The Ruliad and the Vacuum

Stephen Wolfram's Principle of Computational Equivalence posits that simple rules can generate irreducible complexity. Rule 30 is the archetypal example, exhibiting thermodynamic-like chaos from a deterministic seed.

In the APH Cosmology [1], the universe begins in a Sedenionic phase ($\mathbb{S}$) characterized by the failure of the division property ($xy = 0$ for $x, y \neq 0$). We propose that **Rule 30 is a 1+1 dimensional simulation of the Sedenion Vacuum**. The *chaos* observed is not random; it is the deterministic propagation of algebraic defects (zero divisors) in a medium lacking the *stiffness* required to enforce associativity.

2 The Algebraic Formulation of CA

Standard CA theory treats cell values $s_i \in \{0, 1\}$ as bits. We treat them as basis vectors of an algebra $\mathbb{A}_n$. Let the state space be mapped to the Sedenion basis elements:

- $0 \mapsto e_0$ (The Scalar/Vacuum)
- $1 \mapsto e_i$ (The Imaginary/Defect, where $i \in \{1 \dots 15\}$)

The update rule $\Phi(s_{i-1}, s_i, s_{i+1})$ is essentially a non-linear product operator:

$$s_i^{(t+1)} = \mathcal{P}(s_{i-1}^{(t)} \cdot s_i^{(t)} \cdot s_{i+1}^{(t)}) \quad (1)$$

where $\mathcal{P}$ is a projection operator.

3 Rule 30 as the Associator Anomaly

The defining characteristic of Rule 30 is that local information spreads to generate global entropy. In algebra, the failure of locality is measured by the **Associator**:

$$[x, y, z] = (xy)z - x(yz) \quad (2)$$

In the Octonions ($\mathbb{O}$), the associator is non-zero but alternating (Alternative Algebra). In the Sedenions ($\mathbb{S}$), alternativity fails.

Theorem: The chaos of Rule 30 is isomorphic to the non-vanishing of the associator tensor in the Sedenion bulk.

Proof. Let the *left* computation corresponds to $(s_{i-1}s_i)s_{i+1}$ and the *right* computation to $s_{i-1}(s_is_{i+1})$. If the algebra is associative (like $\mathbb{R}$ or $\mathbb{H}$), these pathways interfere destructively, leading to periodic structures (Class 1 or 2). If the algebra is Sedenionic, $[x, y, z] \neq 0$, and the difference propagates as a *noise* term. The central column of Rule 30 is the visual trace of $[x, y, z]$ failing to close. $\square$

4 Geometric Stiffness: The Phase Transition

We now introduce the APH control parameter β (Stiffness). In a physical vacuum, stiffness resists deformation. In a computational vacuum, stiffness resists *Rule Violation*.

We modify the update rule to include a **Koide Filtration Step**:

$$s_i^{(t+1)} = \begin{cases} \Phi_{30}(s_{neighborhood}) & \text{if } |[x, y, z]| < \beta_{crit} \\ 0 & \text{if } |[x, y, z]| \geq \beta_{crit} \end{cases} \quad (3)$$

Regime A ($\beta \rightarrow 0$): The filter is off. The Associator errors propagate unchecked. This is **Rule 30** (The Sedenion Seeds / Primordial Chaos). **Regime B** ($\beta \approx 1.91$): The filter is active. It suppresses the non-associative *noise* while permitting complex structure. This is **Class 4** (The Edge of Chaos / Life). **Regime C** ($\beta \rightarrow \infty$): The filter is rigid. All defects are annihilated immediately. This is **Class 1** (The Frozen Vacuum / Pure Octonion).

5 The Temporal Lifeline

Despite the spatial chaos, Rule 30 exhibits a consistent arrow of time. This validates the **Power Associativity Principle** derived in [1].

$$x^n \cdot x^m = x^{n+m} \quad (4)$$

Even in the Sedenion limits (Rule 30), the algebra of powers remains isomorphic to $\mathbb{C}$. Thus, a unitary time parameter t can be defined even within the chaotic *scrambling* of the cellular lattice.

6 Conclusion: We Live in a Stiff Rule 30

Wolfram asks how the Ruliad (all rules) selects our specific physics. APH provides the answer: *Stiffness Selection*. The universe is not running Rule 30; it is running Rule 30 with a stiffness parameter $\Gamma = \pi$ applied to the vacuum topology. We observe the *Standard Model* because it is the subset of the automaton that survives the geometric stress test. Dark Matter is simply the surface tension of the regions where Rule 30 behavior is still trying to break through.

References

- [1] Schutza, A. M. (2025). *Pregeometric Cosmology: The Sedenion Scream*. Zenodo.
- [2] Reggiani, S. (2024). *The Geometry of Sedenion Zero Divisors*. arXiv:2411.18881.
- [3] Wolfram, S. (2002). *A New Kind of Science*. Wolfram Media.